

EDUCATION

McMaster University	Hamilton, ON, Canada
M.A.Sc. in Computational Science and Engineering, supervised by Dr. H. Mahyar, GPA: 4/4	Sep. 2022–Sep. 2024
Sharif University of Technology - Kish International Campus	Kish, Iran
B.Sc. in Computer Engineering, GPA: 17.78/20 (3.83/4.0)	2017–2022

WORK EXPERIENCE

BASF Canada Inc.	Mississauga, ON, Canada
<i>AI Engineer</i>	Sep. 2024–Present
– Architected a multilingual QA RAG system utilizing FAISS to facilitate global information sharing and streamline internal firm operations. – Deployed scalable customer churn prediction models on Azure Databricks, analyzing large-scale operational datasets to provide predictive intelligence to commercial teams. – Built reproducible data pipelines using Docker and CI/CD best practices, taking ownership of projects from scoping through deployment.	
<i>Applied Machine Learning Intern</i>	Jan. 2023–Sep. 2024
– Collaborated directly with non-technical stakeholders (Business Excellence and Customer Experience teams) across multiple countries to understand requirements and translate business problems into a working Churn Prediction System. – Developed an Accident and Incident Management System with the Generative AI team to automate manual reporting workflows.	
Appraiva	Winnipeg, MB, Canada (Remote)
<i>Machine Learning Engineer Intern</i>	May 2024–Aug. 2024
– Developed AI-powered workflows and a multi-agent real estate evaluation system utilizing LangChain, FastAPI, and Docker to automate property analysis. – Implemented a mixture of experts approach using multi-modal LLMs (OpenAI GPT, Anthropic Claude) to power the backend of an internal web application.	
Vector Institute	Toronto, ON, Canada
<i>Research Analyst</i>	May 2024–Aug. 2024
– Built structured data pipelines and a RAG system using instruct-tuned LLMs to extract and analyze complex datasets from yearly financial reports and websites. – Collaborated with Graph Commons to build a knowledge graph database from collected tabular data, automating manual data entry workflows.	
EthicalAI	Toronto, ON, Canada
<i>Machine Learning Associate</i>	Jan. 2024–May 2024
– Developed forecasting models to improve demand and risk management solutions. – Implemented a novel modular multi-resolution framework to improve existing time-series prediction models.	
Arjan Araye Pouya	Ahwaz, Khuzestan, Iran
<i>Software Engineer</i>	Jun. 2020–Sep. 2022

- Designed and built an internal **salary reporting web application** using **React**, **Django**, and **SQL Server**, supporting thousands of active daily users and streamlining operations.
- Owned the end-to-end development of an external customer web portal providing contract information, leveraging **React** and **PostgreSQL**.
- Set up and maintained a private Git server to improve code quality and development practices for the engineering team.

AWARDS AND HONORS

- Vector Scholarship in Artificial Intelligence Recipient 2022-23
- Ranked within the top 5% among B.Sc. Computer Engineering students
- Distinguished student in Computer Engineering Department at Sharif University of Technology

RESEARCH EXPERIENCE

- Research Assistant, MahyarLab, McMaster University, Hamilton, ON, Canada May 2021–Sep. 2024
Conducted research on scalable and distributed graph representation learning using graph neural networks. Thesis: Aligning knowledge graph embeddings with large language models
- Multi-grid Project, Sharif University of Technology, Kish International Campus 2020
Using traditional methods, it is computationally expensive to solve large sparse linear systems of equations. To address this issue, multi-grid methods are employed. We did research on applying graph representation learning methods to multi-grid solvers.

PUBLICATIONS AND PRE-PRINTS

- [1] **R. Namazi**, E. Ghalebi, S. Williamson, and H. Mahyar, *SMGRL: A Scalable Multi-resolution Graph Representation Learning Framework*, Code: <https://github.com/rezanmz/SMGRL>, 2022. arXiv: 2201.12670 [cs.LG].
- [2] **R. Namazi**, A. Zolanvari, M. Sani, and S. A. A. G. Ghahramani, *GL-Coarsener: A Graph representation learning framework to construct coarse grid hierarchy for AMG solvers*, Code: <https://github.com/rezanmz/GL-Coarsener>, 2020. arXiv: 2011.09994 [math.NA].

PROJECTS

- Home Server Infrastructure
*Architected a self-hosted infrastructure on a Raspberry Pi using Docker Compose and automated CI/CD deployments. Built an internal **Open WebUI** AI assistant enriched with agentic deep research pipelines and smart context management to automate workflows. Code available at <https://github.com/rezanmz/home-server>*
- Graph Neural Network Architecture Search
This project's purpose is to optimise hyper-parameter tuning of graph neural networks in a large search space. To speed up the process, we first search a low-resolution version of the training graph, then increase the network's quality on a zoomed-in version of the same graph. The search is done using Optuna. Code available at <https://github.com/rezanmz/GNN-NAS>
- Molecule generation using Graph Convolutional Network (GCN)
Using GCN in a GAN-like setting, I trained a generative model that outputs the structure of molecules similar to the seen training data. Code available at <https://github.com/rezanmz/MolGenerator>
- K-Means clustering from scratch with automatic selection of K
I have implemented the K-Means clustering algorithm from scratch in C++. I also utilized the Elbow method to automatically select the best number of clusters. Code available at <https://github.com/rezanmz/kmeans-cpp>

- Modeling Epidemics
In this project I tried to analyze an epidemic with infection rate α and recovery rate β in an SIS (Susceptible - Infected - Susceptible) model.
- A naive implementation of a two-grid multigrid algorithm
*Solve very large sparse linear systems using a Python and C++ implementation of the multigrid algorithm.
Python code available at <https://github.com/rezanmz/AMG>
C++ code available at <https://github.com/rezanmz/multigrid>*

TEACHING EXPERIENCE

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| <ul style="list-style-type: none"> • Teaching Assistant at McMaster University
<i>Introduction to Computational Natural Language Processing (SEP 775)
Holding office hours, designing and grading assignments and final project</i> | Winter 2024 |
| <ul style="list-style-type: none"> • Teaching Assistant at McMaster University
<i>Introduction to Mathematical Scientific Computation (MATH 1MP3)
Grading</i> | Winter 2024 |
| <ul style="list-style-type: none"> • Teaching Assistant at McMaster University
<i>Engineering Mathematics (Math 1ZC3)
MATLAB Labs</i> | Fall 2023 |
| <ul style="list-style-type: none"> • Teaching Assistant at McMaster University
<i>Cyber Physical Systems (SEP 769)
Holding office hours, designing and grading assignments and final project</i> | Winter 2023 |
| <ul style="list-style-type: none"> • Teaching Assistant at McMaster University
<i>Deep Learning (SEP 740)
Holding office hours, designing and grading assignments and final project</i> | Winter 2023 |
| <ul style="list-style-type: none"> • Teaching Assistant at McMaster University
<i>Artificial Intelligence and Machine Learning (SMRTTECH 4AI3)
Holding office hours, grading assignments and final project</i> | Fall 2022 |

SKILLS

- **Programming Languages:** Python, C++, SQL
- **Cloud & MLOps:** Azure, Databricks, AWS, Docker, Kubernetes, Linux, CI/CD, git
- **Frameworks & Libraries:** PyTorch, TensorFlow, FastAPI, Flask, HuggingFace, LangChain, Weights & Biases, Pinecone

LANGUAGES

- **Persian:** Native
- **English:** Professional Proficiency

REFERENCES

Available upon request